

PROBLEM STATEMENT #52 | Media, Events & Community

Community Tool & Equipment Library

1. Problem Context & Overview

A neighborhood resource-sharing library where members check out power tools, camping gear, and lawnmowers, track deposit calculations, manage return dates, and log damage assessments.

Target Stakeholders / Actors: *Community Member, Library Custodian*

2. Sample Functional Requirement (FR) Guideline

Sample Requirement: FR-001 [Priority: High]

Description: The system shall calculate required security deposits based on tool category, enforce loan duration limits, and calculate late return fees.

Sample Acceptance Criteria: Pass: Tool reservation confirmed with deposit hold; Fail: Member with overdue tools borrows additional high-value equipment.

3. Sample Non-Functional Requirement (NFR) Guideline

Sample Requirement: NFR-001 [Type: Performance & Security]

Description: The tool inventory search must display real-time physical availability across neighborhood depot lockers.

Sample Acceptance Criteria: Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

Deliverables to Submit: [All through a GitHub Repo in your account]

- **Complete Requirements Table:**

- >> Exactly 5 Functional Requirements (FR-001 to FR-005)
- >> 2 Non-Functional Requirements (NFR-001 & NFR-002) with ID, Type, Description, Priority, Acceptance Criteria, and Rationale.

- **UML Use-Case Diagram:**

- >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.

- **Use-Case Flow Specification:**

- >> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario
- >> One Alternate Flow.